



# CODIORA

OUR CODE BUILDS YOUR VISION

## WEEK 4 TASKS

### 1. Frontend Development

#### Task Title:

Interactive Portfolio with API Integration

#### Objective

In the previous weeks, you built and enhanced your portfolio website with modern UI components. This week, your goal is to connect your frontend with live backend APIs, making your portfolio dynamic instead of using static data.

#### Requirements

##### 1. Integrate Portfolio APIs

Replace all hardcoded data by fetching information from backend APIs.

The following sections should retrieve data dynamically:

- Profile Information
- Skills
- Projects
- Contact Information

## **2. Dynamic Project Management**

Display projects dynamically from the API.

Each project should include:

- Project Image
- Project Name
- Description
- Technologies
- Category
- GitHub Link
- Live Demo Link

## **3. Loading & Error States**

Implement:

- Loading Spinner
- Skeleton Loaders
- Error Messages
- Empty State UI

## **4. Search & Filter Improvements**

Allow users to:

- Search Projects
- Filter by Category
- Filter by Technology

## **5. Reusable Components**

Refactor the project into reusable components.

Examples:

- Project Card
- Button
- Navbar
- Footer
- Modal
- Skill Card

## **6. Responsive Optimization**

Ensure the application works perfectly on:

- Mobile
- Tablet
- Desktop

## **Recommended Technologies**

- React.js
- Next.js
- Vue.js
- Tailwind CSS
- Fetch API

## **Evaluation Criteria**

- API Integration
- Component Reusability
- Code Organization
- User Experience
- Responsive Design

## **2. Backend Development**

### **Task Title:**

Professional Backend Services & API Optimization

### **Objective**

In previous weeks, you developed portfolio APIs. This week, enhance the backend by improving performance, security, and API architecture to match real-world development practices.

### **Requirements**

#### **1. Image Storage System**

Implement image upload functionality for:

- Profile Images
- Project Images

Store images securely and return accessible URLs.

#### **2. API Validation**

Validate every incoming request.

Examples:

- Required Fields
- Email Validation
- Password Rules
- Data Types

#### **3. Advanced Search APIs**

Support:

- Keyword Search

- Category Filtering
- Technology Filtering
- Pagination

#### **4. Centralized Error Handling**

Create a professional error-handling middleware.

Return meaningful HTTP responses.

#### **5. API Documentation**

Document every endpoint with:

- Request Body
- Response
- Error Codes

#### **6. Performance Optimization**

Improve:

- Database Queries
- API Response Time
- Code Structure

#### **Recommended Technologies**

- Node.js
- Express.js
- MongoDB
- PostgreSQL
- JWT

- Multer
- Swagger

### **Evaluation Criteria**

- Security
  - Performance
  - Documentation
  - API Quality
  - Code Structure
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### **3. Full Stack Development**

#### **Task Title:**

Complete Portfolio Management System

#### **Objective**

Your goal is to fully integrate the frontend and backend into one professional Portfolio Management System.

#### **Requirements**

##### **1. Complete API Integration**

Connect every frontend page with backend APIs.

##### **2. Dashboard Enhancements**

Implement:

- Dashboard Statistics
- Recent Activity
- Notifications

##### **3. Project Management**

Allow users to:

- Create Projects
- Edit Projects
- Delete Projects
- Upload Project Images

##### **4. Profile Management**

Allow users to:

- Update Profile
- Change Password
- Upload Profile Picture

## **5. Authentication Protection**

Protect private routes using authentication.

Unauthorized users should not access dashboard pages.

## **6. Dashboard Optimization**

Improve:

- Performance
- Navigation
- Responsive Layout

## **Recommended Technologies**

- MERN Stack
- Next.js
- Laravel
- Django
- PostgreSQL

## **Evaluation Criteria**

- Full Stack Integration
  - Dashboard Functionality
  - Authentication
  - Database Design
  - User Experience
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## **4. Mobile App Development**

### **Task Title:**

Portfolio Mobile App with Backend Integration

### **Objective**

Transform your mobile portfolio into a real application by integrating backend APIs and improving the user experience.

### **Requirements**

#### **1. API Integration**

Fetch data dynamically for:

- Profile
- Skills
- Projects
- Contact Details

#### **2. Authentication**

Implement:

- Login
- Logout
- Secure Session Management

#### **3. Project Details**

Display complete project information with images.

#### **4. Search & Filter**

Enable users to:

- Search Projects

- Filter Categories

## **5. Image Upload**

Allow users to update their profile image.

## **6. Performance Optimization**

Improve:

- Navigation
- API Calls
- State Management

## **Recommended Technologies**

- Flutter
- React Native
- Firebase
- REST API
- Riverpod
- GetX

## **Evaluation Criteria**

- API Integration
- Performance
- Navigation
- UI Design
- Code Quality

## 5. Graphic Design

### Task Title:

Professional Digital Brand Identity

### Objective

Expand your personal branding into a complete professional identity that can be used for freelance platforms, portfolios, and social media marketing.

### Requirements

#### 1. Portfolio UI Design

Design complete desktop and mobile portfolio screens.

#### 2. Freelancer Profile Kit

Design assets for:

- Fiverr Profile Banner
- Upwork Cover
- LinkedIn Banner

#### 3. Social Media Campaign

Create a complete campaign including:

- Instagram Posts
- LinkedIn Posts
- Story Templates

#### 4. Presentation Template

Design a professional presentation theme containing:

- Cover Slide
- Content Slides
- Closing Slide

## **5. UI Design System**

Prepare:

- Typography
- Buttons
- Cards
- Icons
- Color Palette

## **6. Design Mockups**

Present your work using realistic mockups.

### **Recommended Tools**

- Figma
- Adobe Photoshop
- Adobe Illustrator
- Canva

### **Evaluation Criteria**

- Design Consistency
  - Branding
  - Creativity
  - Professional Quality
  - Presentation
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## **6. Data Analysis**

### **Task Title:**

Business Intelligence Portfolio Dashboard

### **Objective**

Build an advanced Business Intelligence dashboard that transforms raw portfolio data into actionable insights using interactive analytics and professional reporting.

### **Requirements**

#### **1. Data Transformation**

Prepare the dataset by:

- Cleaning Data
- Removing Duplicates
- Standardizing Formats
- Creating Calculated Fields

#### **2. Interactive Dashboard**

Develop dashboards with:

- Filters
- Slicers
- Drill-Down Features
- KPI Cards

#### **3. Advanced Analytics**

Analyze:

- Technology Usage Trends

- Project Performance
- Skill Distribution
- Completion Status

#### **4. Business Insights**

Generate insights such as:

- Most Popular Technologies
- Fastest Growing Skill Categories
- Project Completion Trends
- Performance Recommendations

#### **5. Automated Reports**

Create a professional report summarizing:

- Dashboard Findings
- Key Metrics
- Recommendations
- Future Improvements

#### **6. Dashboard Presentation**

Present your analysis using interactive visualizations and a clean, business-focused layout.

#### **Recommended Technologies**

- Power BI
- Tableau
- Excel
- Python

- SQL
- Pandas

### **Evaluation Criteria**

- Dashboard Design
  - Data Accuracy
  - Business Insights
  - Visualization Quality
  - Reporting Skills
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## Week 4 Outcome

By the end of **Week 4**, interns should have transformed their projects into **real-world, production-style applications**.

- **Frontend Developers** will have a dynamic portfolio integrated with backend APIs.
- **Backend Developers** will deliver secure, optimized, and well-documented REST APIs.
- **Full Stack Developers** will complete an end-to-end Portfolio Management System with authentication and dashboard functionality.
- **Mobile App Developers** will build a fully functional portfolio app powered by live APIs.
- **Graphic Designers** will produce a comprehensive digital branding system suitable for professional platforms.
- **Data Analysts** will create an interactive Business Intelligence dashboard that communicates meaningful insights through professional visualizations.



## **LinkedIn Submission**

### **After completing your task:**

1. Share your work on LinkedIn.
2. Include screenshots of your progress.
3. Briefly explain what you built and what you learned.
4. Submit your LinkedIn post link through the provided **Week 4** Google Form.

### **Submission Guidelines**

- Use any technology of your choice.
- Focus on quality rather than quantity.
- Keep your work organized and professional.
- Creativity and additional features are encouraged but not mandatory.

### **Submission Deadline**

Submit your work before the assigned deadline.

### **Best of Luck!**

Codiora House (Private) Limited

*Our Code Builds Your Vision*